

Unit 3: Spectroscopic Techniques & Applications

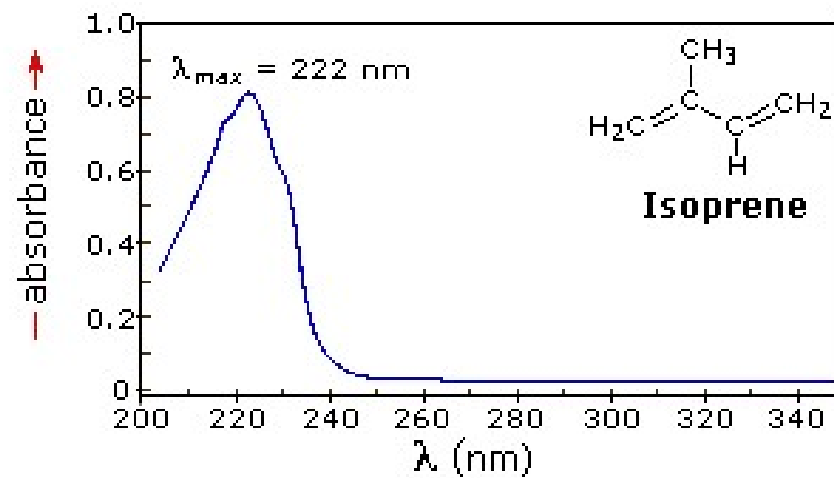
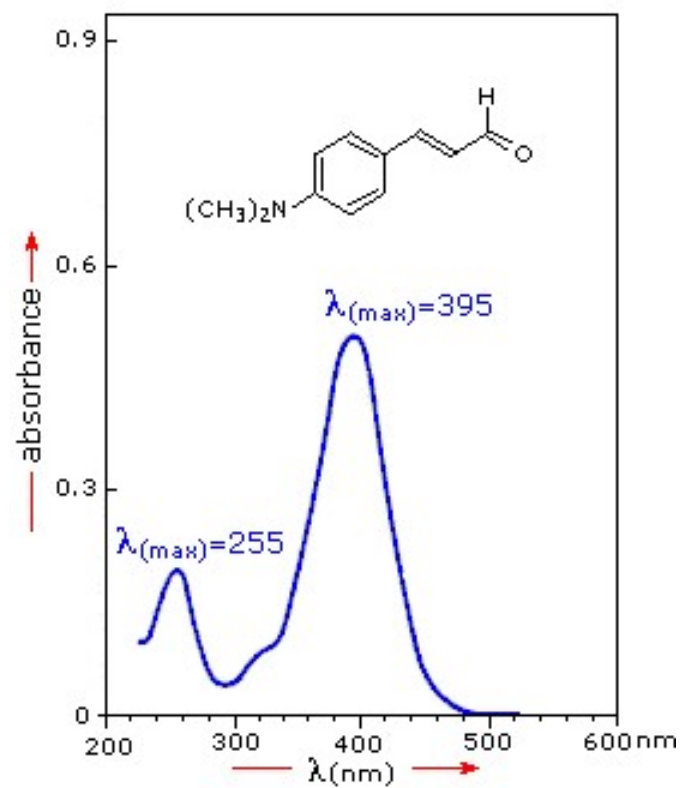
Spectroscopy

- It is a branch of science that deals with the interaction of matter with light or electromagnetic radiation.

OR

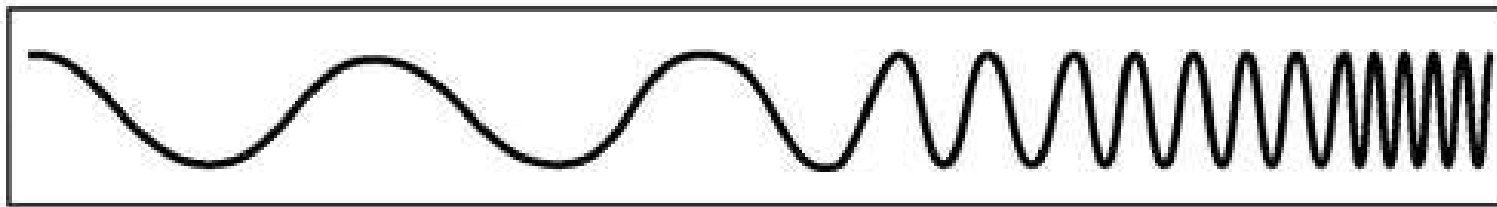
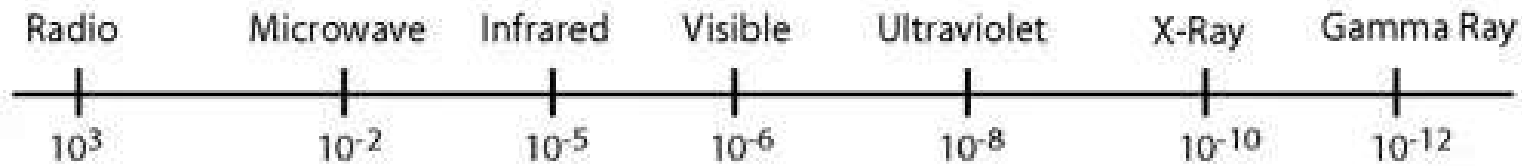
- It is a branch of science in which electromagnetic radiation of particular wavelength or range of wavelength is used for qualitative.
- The matter can be **atoms, molecules**, or **ions**.
- The measurement of such **signals as a function of wavelength** results in the collection of a **spectrum**.
 - It is the most powerful tool available for the study of **atomic and molecular structures**.
 - It is used in the analysis of a wide range of samples.

Spectrum:

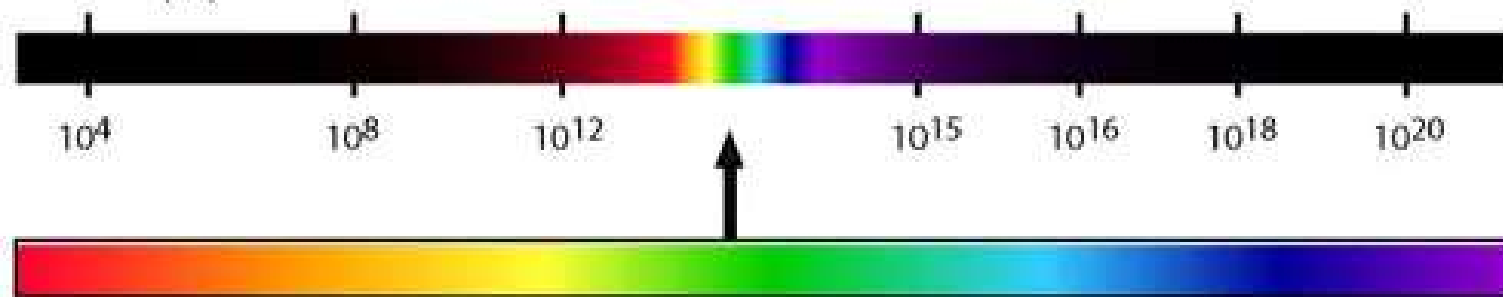


THE ELECTRO MAGNETIC SPECTRUM

Wavelength
(metres)



Frequency
(Hz)



$$c = \lambda \nu$$

speed of light wavelength frequency

$$E = h \nu$$

energy planck's constant frequency

$$E = \frac{hc}{\lambda}$$

energy planck's constant speed of light wavelength

Characterization of EMR

Wavelength (λ) : It is the distance between two consecutive peaks of a wave (crests). Unit = m.

Frequency (ν): It is the number of waves that are formed in a given length of time. Unit = number of wave cycles per second or hertz (Hz).

A short wavelength means that the frequency will be higher and a longer wavelength has a lower frequency.

Wavenumber ($\bar{\nu}$) : It is the number of waves per unit distance. Unit = cm^{-1} $\bar{\nu} = 1/\lambda$

Energy of EMR (E): Electromagnetic radiations consists of particles having small packets of energies called quanta or photons. Photons possess the characteristic of wave and travel with the speed of light. The amount of energy corresponding to one photon is expressed by Planck's equation as

$$E = h\nu \text{ or } E = \frac{hc}{\lambda}$$

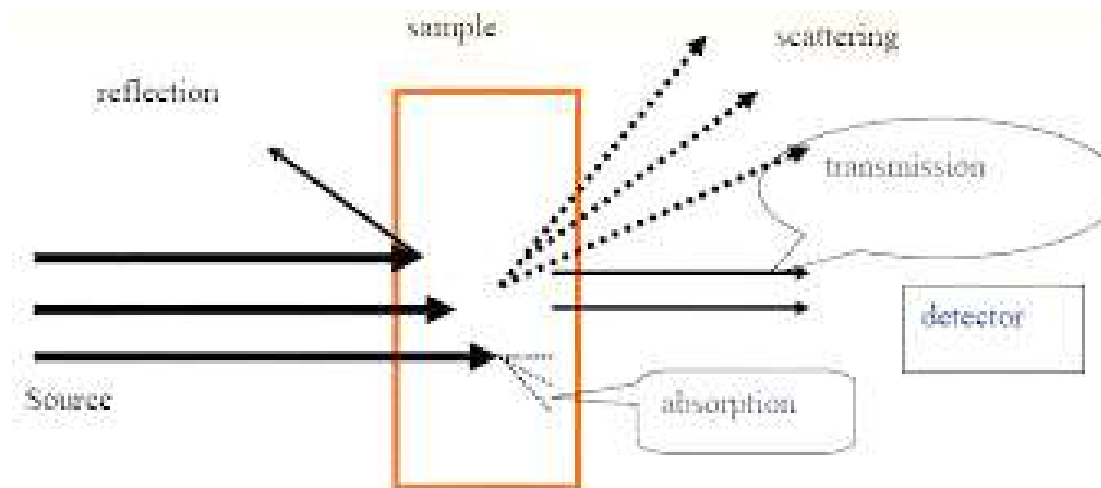
where h = Planck's constant ($6.62 \times 10^{-34} \text{ Js}$)

ν - frequency in Hz

λ - wavelength in cm/m.

Interactions of EMR with Matter

- Absorption – no re-emission energy dissipated as heat
- Absorption/Re-emission – if the **same wavelength** is emitted then it is called light scattering
- Absorption/Re-emission – if **different wavelength** then it is called fluorescence or phosphorescence

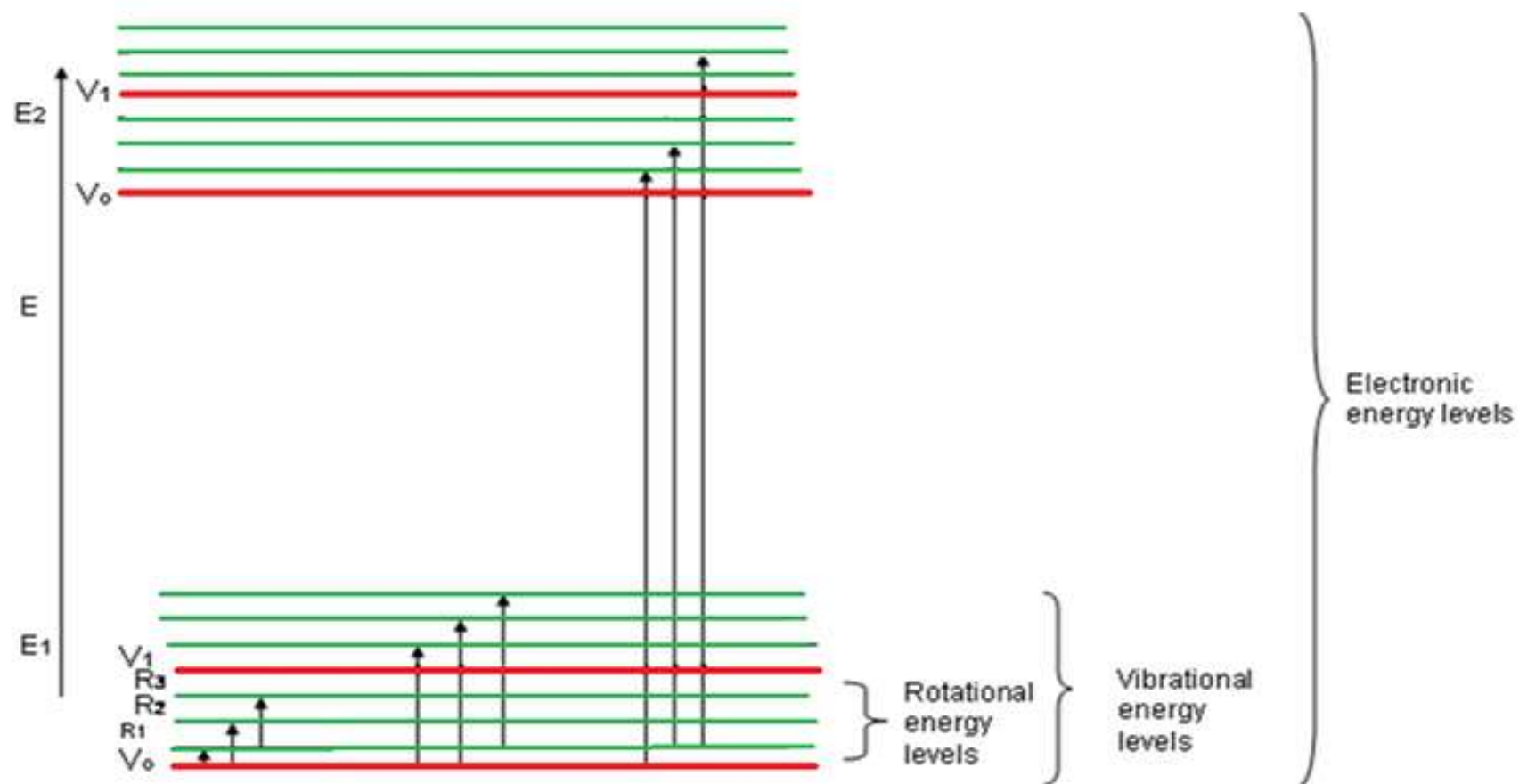


Interaction of EMR with matter

- EMR interacts with matter only when the matter has some electric and magnetic effect and are influenced by the electric and magnetic components of the EM radiation.
- The net change in the electric/magnetic dipole moment in the molecule or nuclear spin, interact with the magnetic/electrical component of the EMR by either absorption or emission of the EMR.
- Total energy of molecules = Translational + rotational + vibrational + electronic
- Absorption or emission of EMR causes a change in any of these types of energies.

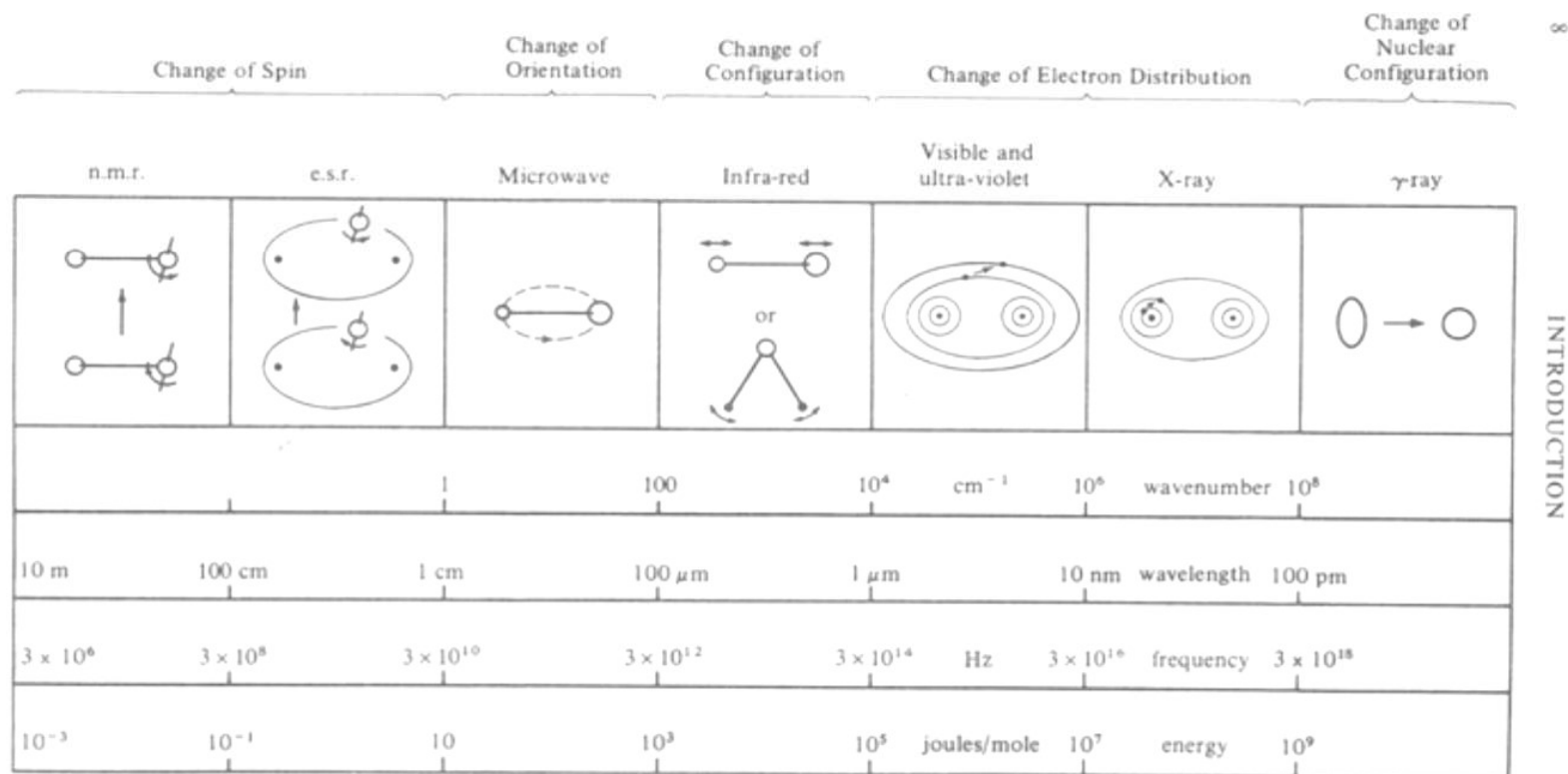
In molecular spectroscopy, we measure the change in these energy states.

1. **Translational energy** – It is due to the overall movement of the molecule. Energy levels are not quantized . Hence no spectroscopy.
2. **Rotational energy** – It is due to spinning of molecules about the axis passing through the centre of gravity - Rotational Levels are quantized – **Rotational spectroscopy (Microwave spectroscopy)**
3. **Vibrational energy** – It is due to vibrations in molecules – Vibrational Levels are Quantized – **IR Spectroscopy (Vibrational spectroscopy).**
4. **Electronic energy** – Consists of electronic levels which are quantized – **UV/visible spectroscopy (Electronic spectroscopy)**



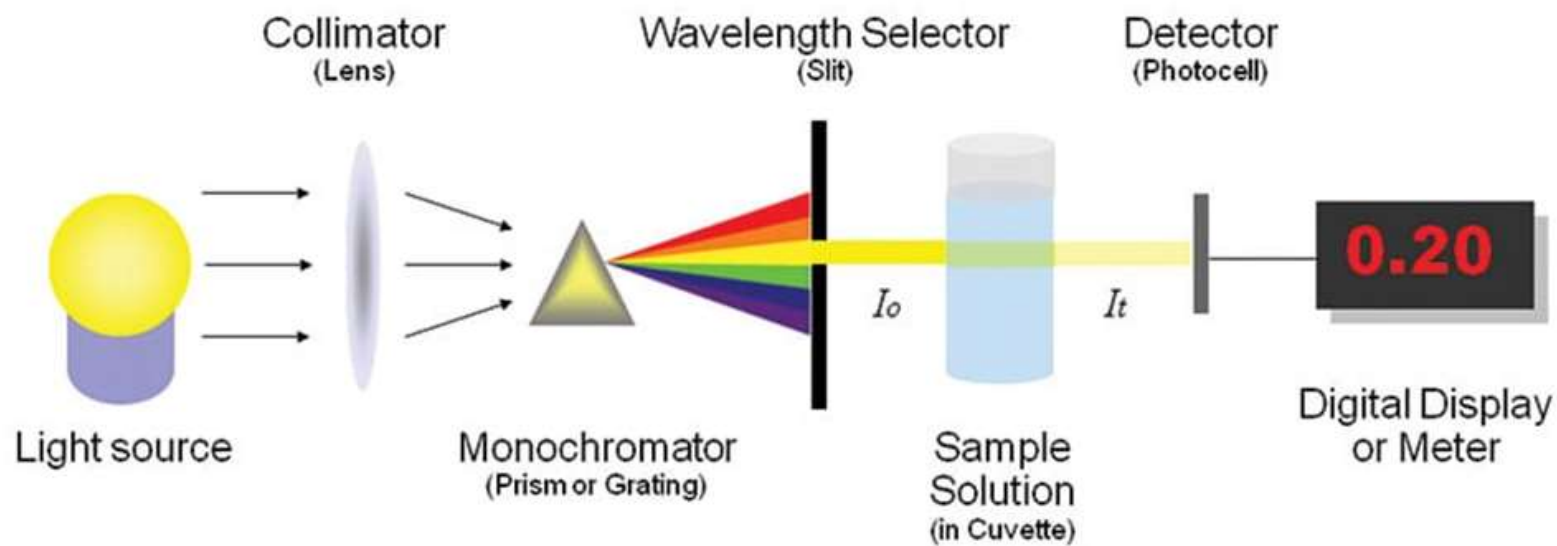
The type of EMR depends on wavelength, frequency, energy, etc

Absorption or emission of specific types of EMR cause changes in different kinds of atomic or molecular energy



Principle of Spectroscopy:

- It is based on the measurement of the spectrum of a sample containing atoms/molecules



Principles of Spectroscopy

1. Absorption Spectroscopy:

- An analytical technique which concerns with the measurement of absorption of electromagnetic radiation.
- e.g. UV (185 - 400 nm) / Visible (400 - 800 nm) Spectroscopy, IR Spectroscopy (0.76 - 15 μm)

2. Emission Spectroscopy:

- An analytical technique in which emission (of a particle or radiation) is dispersed according to some property of the emission & the amount of dispersion is measured.
- e.g. Mass Spectroscopy

Why we use spectroscopy

- **Applications**

- Spectroscopy is used as a tool for studying the structures of atoms and molecules. The large number of wavelengths emitted by these systems makes it possible to investigate their structures in detail.
- Spectroscopy also provides a precise analytical method for finding the constituents in material having unknown chemical composition.
- In a typical spectroscopic analysis, a concentration of a few parts per million of a trace element in a material can be detected through its emission spectrum

UV-VISIBLE SPECTROSCOPY

❖ *UV spectroscopy* $\longrightarrow$ *Electronic Spectroscopy*

❖ It involves the *promotion of electrons* from ground state to excited (higher energy) in a molecule.

❖ UV Spectroscopy deals with the measurement of energy absorbed when electrons are promoted to higher energy levels.

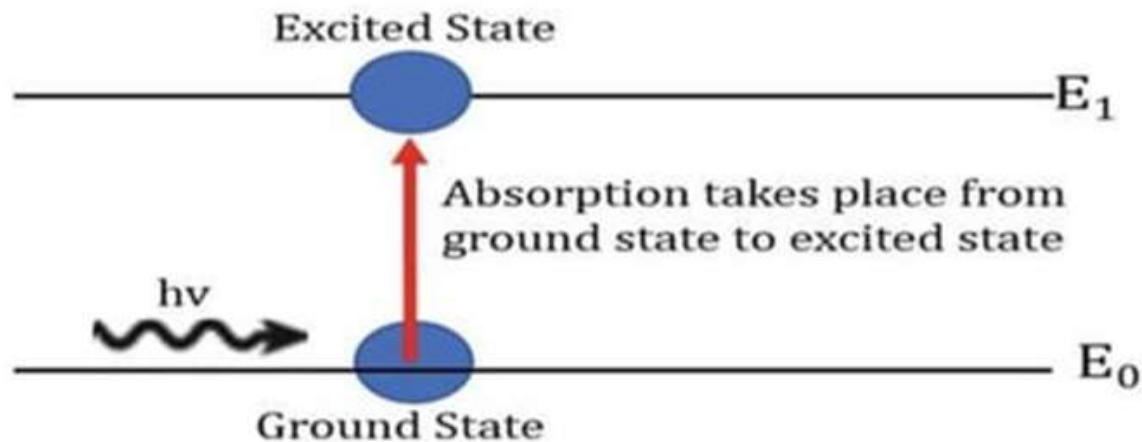
❖ UV spectrum is simply a *plot of wavelength of light absorbed Vs the absorption intensity (absorbance or transmittance)*

❖ since UV radiation associated with *larger amount of energy*, they are capable of electronic transitions and induce transitions in electronic, vibrational and rotational energy levels of a molecule.

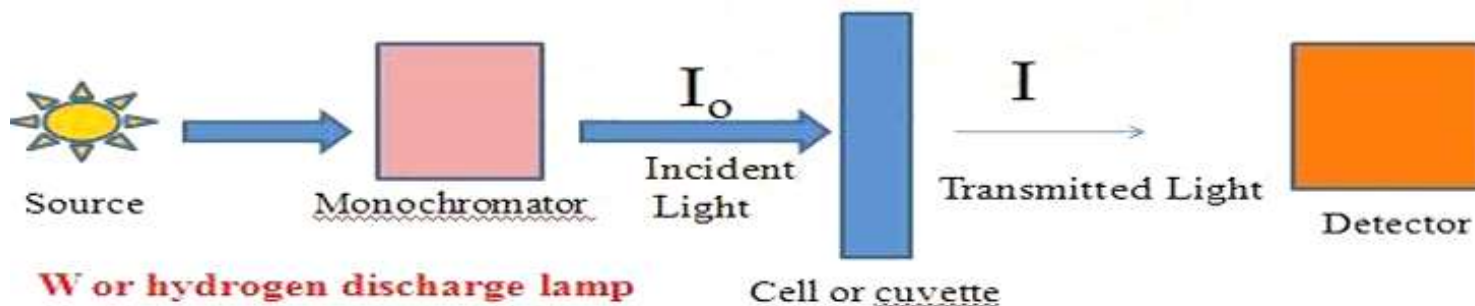
RANGE : 200 $\longrightarrow$ 800 nm

PRINCIPLE

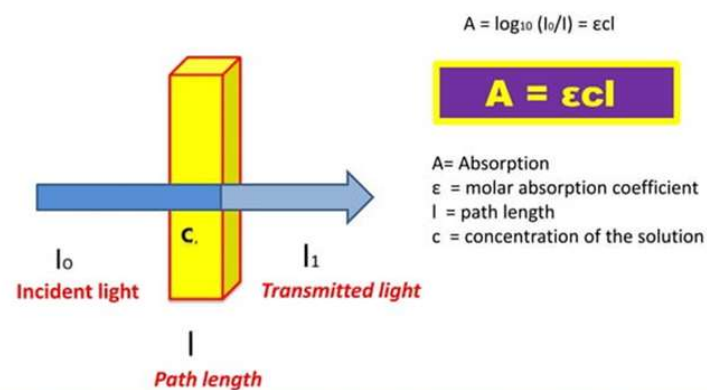
- The principle is based on the *absorption of ultraviolet light* or visible light by chemical compounds, which results in the production of spectra
- While interaction between light and matter, the matter absorbs light, it *undergoes excitation and de-excitation*, resulting in the production of a spectrum.
- while absorption the *electrons from ground state excited to higher energy state*.
- the energies of the GS and the ES of the electron is always equal to the amount of UV visible radiation absorbed by it



UV-Visible spectroscopy – Instrumentation



BEER – LAMBERTS LAW



The amount of energy absorbed or transmitted by a solution is proportional to the solution's Molar absorptivity and the concentration of the solute.

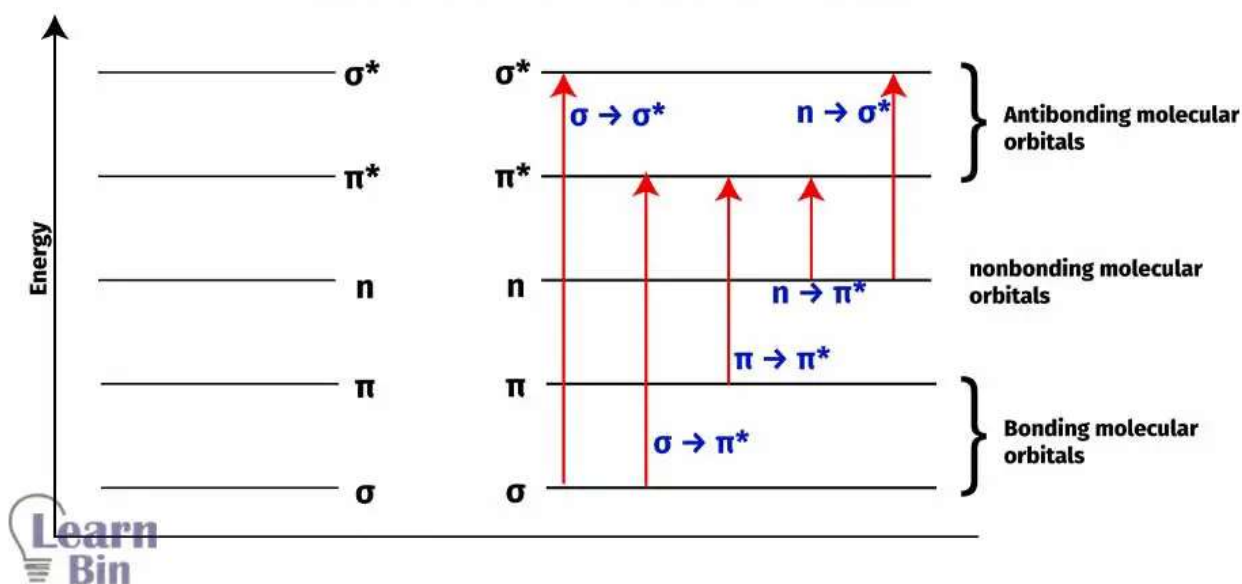
UV-Visible spectroscopy – Instrumentation

- **Sources:** Commonly used sources of UV radiation are the hydrogen lamp and the deuterium lamp. Tungsten filament lamp is used for the visible range.
- **Filters and Monochromators:** Tinted glass filters are used to produce monochromatic radiations. Filters resolve polychromatic light into a relatively wide bandwidth to produce monochromatic radiations. A monochromator is a device that resolves polychromatic radiation into its individual wavelength and isolates them into very narrow bands.
- **Prism:** A prism disperses polychromatic light from a source into its constituent wavelengths. Cornu quartz and Littrow prisms are used. Glass prisms are used for visible range while silica, fused silica or quartz prisms are used for the UV range.
- **Sample Holder (Cells / Cuvette):** The sample container should be transparent to the UV-visible radiation. Cuvettes are made of ordinary glass or quartz. Fused silica cells are used for the UV range. The path length of these cuvettes is usually 1 cm.
- **Solvents:** A solvent is selected in such a way that it does not absorb in the UV-visible range. The solvents that are frequently used are water, methanol, ethanol, hexane, chloroform etc.
- **Detectors:** The commonly used detectors are photocells, phototubes and photomultiplier tubes.
- **Amplification and Read out:** The transmitted radiation is converted into electrical signal and the electrical signals are interpreted using ammeters, amplifiers and potentiometers.

ELECTRONIC TRANSITIONS

- According to the MOT, when a molecule is excited by the absorption of energy, its electrons are promoted from bonding to antibonding orbital
- All the *saturated bonds are σ – bonds* its corresponding *antiorbital is σ^**
- All the *unsaturated bonds are π -bonds*, its corresponding *antiorbital is π^** .
- All the nonbonding unshared pair of electrons or lone pair of electrons are present in n-orbitals (non bonding orbitals).

Electronic energy levels and transitions



$\sigma \longrightarrow \sigma^*$ Transition

- It is the transition of an electron from **bonding σ -orbital to its antibonding orbital** .
- It is a **high energy process** because σ -bonds are **very strong (stable)**.
- Thus, these transitions **do not occur in normal UV-regions** (200-800nm)
- so, $\sigma \longrightarrow \sigma^*$ transition is less informative in UV.
- eg., saturated hydrocarbons like methane , propane etc ., absorption occurs near 150m μ .
- Thus, and **evacuated spectrophotometer** is used for studying such high energy transitions.
- Below 200m μ , since oxygen begins to absorb strongly . Therefore, the lower region is evacuated with nitrogen air.
- this region is called **Vacuum region**.
- The spins of electrons is not change from GS to ES.
- It occurs only singlet to singlet and triplet to triplet.

$n \longrightarrow \sigma^*$ Transition

- The transition of an electron from a *nonbonding orbital to an antibonding sigma orbital* is called $n \longrightarrow \sigma^*$ transition.
- It takes place in *saturated compounds containing one hetero atom* with unshared pair of electrons.
- Eg., saturated halides, alcohols, ethers, aldehydes, ketones, amines etc.,
- it requires less energy than $\sigma \longrightarrow \sigma^*$ transition.

water	167m μ
Methyl alcohol	174m μ

$\pi \longrightarrow \pi^*$ Transition

- It is the transition of an electron from a *π -bonding orbital to a π antibonding orbital*.
- it occurs in *unsaturated centers of the molecule*.
i.e., compounds containing double bonds, triple bonds and also in aromatic compounds.
- These transitions require lower energy than in $n \longrightarrow \sigma^*$ transitions. Therefore, it *occurs at longer wavelength*.
- eg., alkenes, alkynes, carbonyl compounds, cyanides, azo compounds.
- In alkenes, absorption occurs at 170-190m μ

$n \longrightarrow \pi^*$ Transition

- The transition of an electron from nonbonding orbital to antibonding π orbital is designated $n \longrightarrow \pi^*$ transitions.
- It requires **lowest energy than the other three transitions**. Therefore it occurs at longer wavelength.
- saturated aldehydes show both the types of transitions.

low energy	$n \longrightarrow \pi^*$	290m μ
high energy	$\pi \longrightarrow \pi^*$	180m μ
- absorption occurring at **lower wavelength is more intense**.
- In simple cases, since the extinction coefficient of $n \longrightarrow \pi^*$ is low than $\pi \longrightarrow \pi^*$.

Terms used in UV-visible spectroscopy

Chromophore

The part of a molecule responsible for imparting color, are called as chromospheres.

OR

The functional groups containing multiple bonds capable of absorbing radiations above 200 nm due to $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$ transitions.

e.g. NO_2 , N=O , C=O , C=N , $\text{C}\equiv\text{N}$, C=C , C=S , etc

Auxochrome

The functional groups attached to a chromophore which modifies the ability of the chromophore to absorb light, altering the wavelength or intensity of absorption.

OR

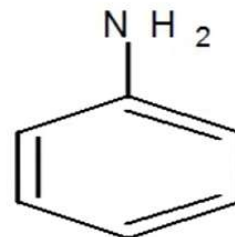
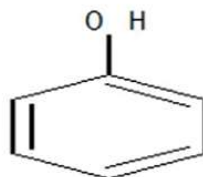
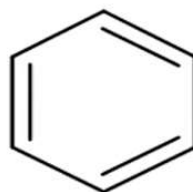
The functional group with non-bonding electrons that does not absorb radiation in near UV region but when attached to a chromophore alters the wavelength & intensity of absorption.

e.g.

Benzene $\lambda_{\text{max}} = 255 \text{ nm}$

Phenol $\lambda_{\text{max}} = 270 \text{ nm}$

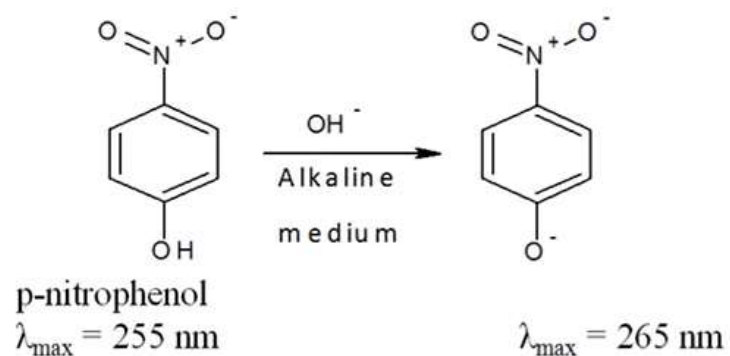
Aniline $\lambda_{\text{max}} = 280 \text{ nm}$



1

• Bathochromic Shift (Red Shift)

- In alkaline medium, p-nitrophenol shows red shift. Because negatively charged oxygen delocalizes more effectively than the unshared pair of electron.



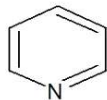
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• Hypsochromic Shift (Blue Shift)

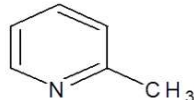
- When absorption maxima (λ_{max}) of a compound shifts to shorter wavelength, it is known as hypsochromic shift or blue shift.
- The effect is due to presence of an group causes removal of conjugation or by the change of solvent.

3 • Hyperchromic Effect

- When absorption intensity (ϵ) of a compound is increased, it is known as hyperchromic shift.
- If auxochrome introduces to the compound, the intensity of absorption increases.



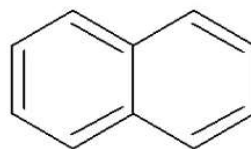
Pyridine
 $\lambda_{\max} = 257 \text{ nm}$
 $\epsilon = 2750$



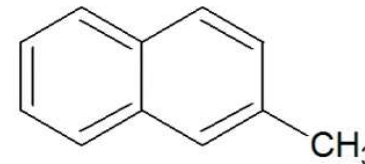
2methylpyridine
 $\lambda_{\max} = 260 \text{ nm}$
 $\epsilon = 3560$

4 • Hypochromic Effect

- When absorption intensity (ϵ) of a compound is decreased, it is known as hypochromic shift.

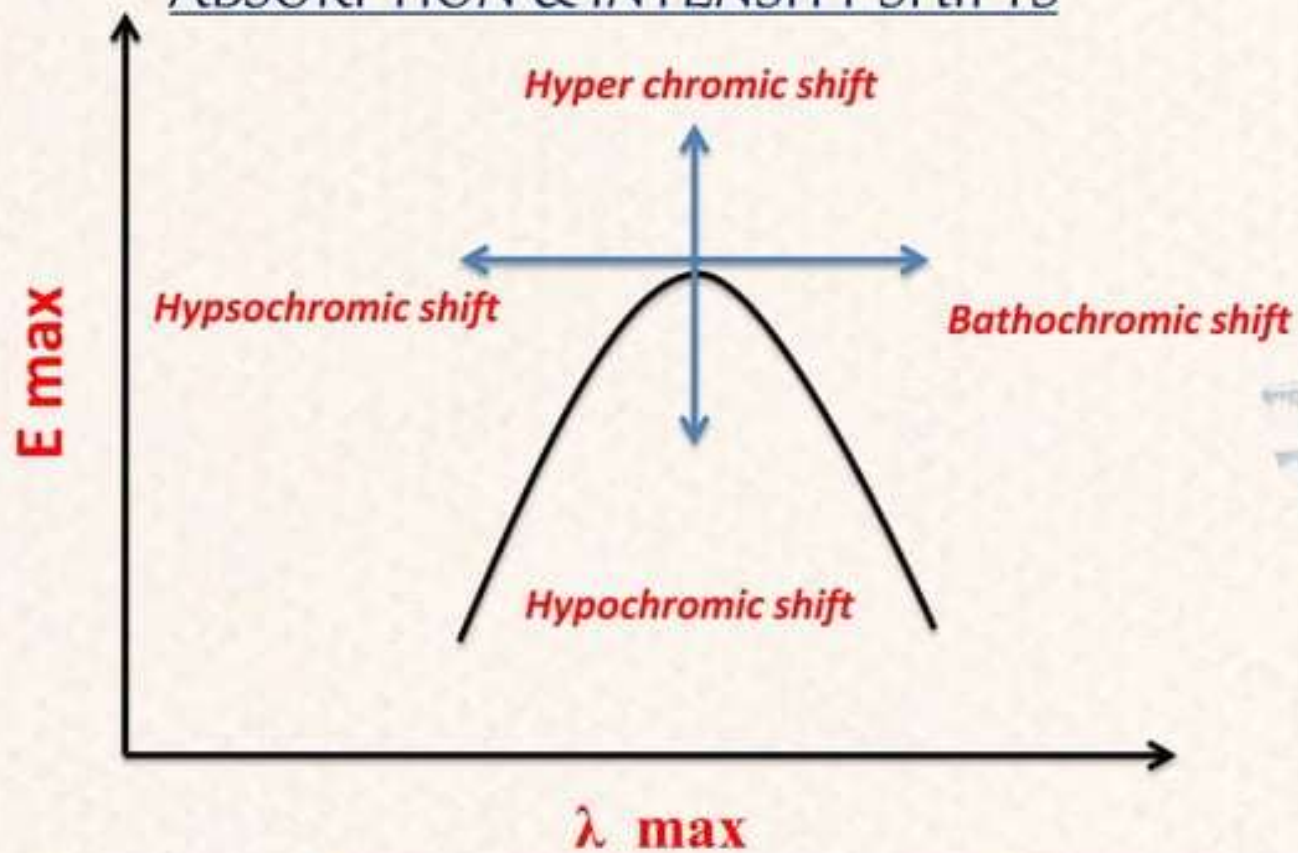


Naphthalene



2-methyl naphthalene

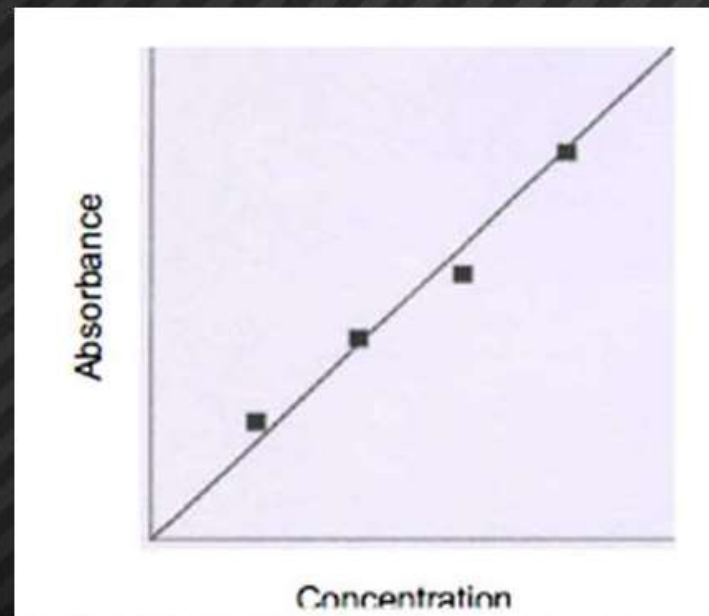
ABSORPTION & INTENSITY SHIFTS



APPLICATIONS OF UV SPECTROSCOPY

1. Concentration measurement

- Prepare samples
- Make series of standard solutions of known concentrations [4]

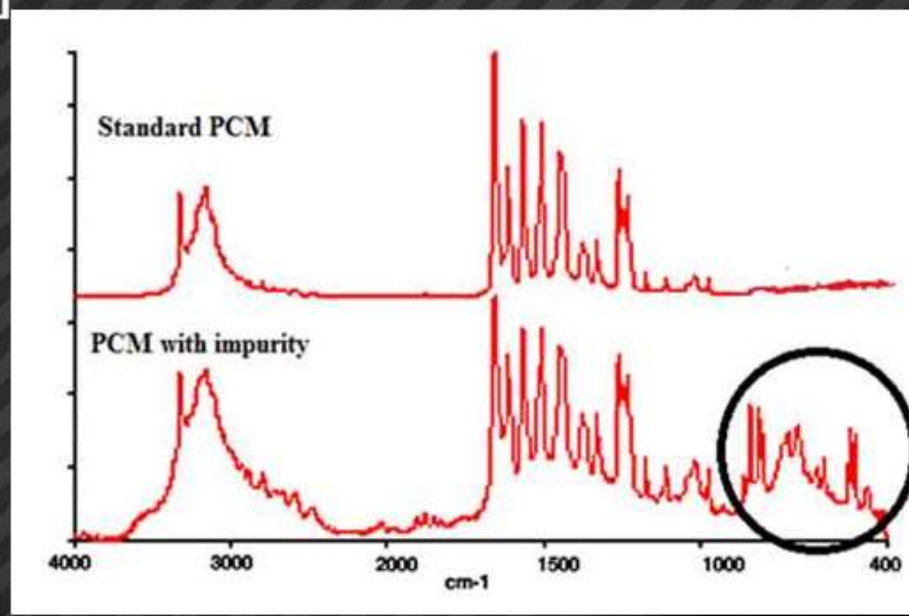


APPLICATIONS OF UV SPECTROSCOPY

2. Detection of Impurities

- UV absorption spectroscopy is one of the best methods for determination of impurities in organic molecules. [7]

Additional peaks can be observed due to impurities in the sample and it can be compared with that of standard raw material.



3 Structure elucidation of organic compounds.

- UV spectroscopy is useful in the structure elucidation of organic molecules, the presence or absence of unsaturation, the presence of hetero atoms.
- From the location of peaks and combination of peaks, it can be concluded that whether the compound is saturated or unsaturated, hetero atoms are present or not etc.

4. QUALITATIVE ANALYSIS

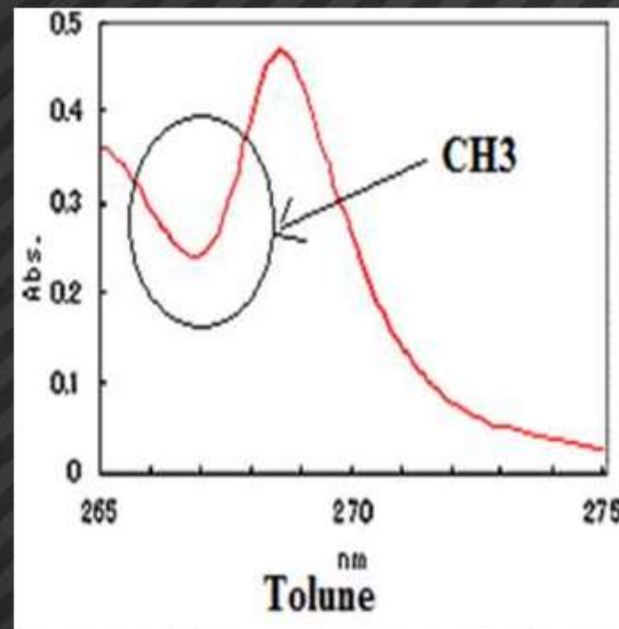
- UV absorption spectroscopy can characterize those types of compounds which absorb UV radiation. Identification is done by comparing the absorption spectrum with the spectra of known compounds.

5. CHEMICAL KINETICS

- Kinetics of reaction can also be studied using UV spectroscopy. The UV radiation is passed through the reaction cell and the absorbance changes can be observed.

5. Detection of Functional Groups

- Absence of a band at particular wavelength regarded as an evidence for absence of particular group [5]



Relation between Absorbance A (OD) and % T.

Beers law:

Transmittance (T) = I/I_0

% T = 100 T

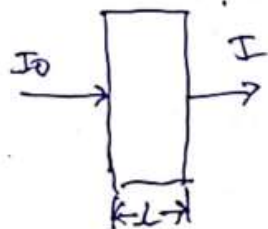
Absorbance (A):

$$A = \log_{10} I_0/I$$

$$A = \log_{10} 1/T$$

$$A = \log_{10} 100/\%T$$

$$A = 2 - \log_{10} (\%T) \checkmark$$



$$A = \frac{I_0}{I}$$

$$A = \log_{10} \left(\frac{1}{T} \right)$$

$$A = \log_{10} \left(\frac{100}{T[\%]} \right) = 2 - \log T[\%]$$

or $A \propto c$ — (1)

Absorbance (A) is defined $A = \log_{10} \left(\frac{I_0}{I} \right)$ — (2)

As (2) can be expressed as $A \propto l$ — (3)

Combining (1) & (3) $A \propto cl$

$$A = \log_{10} \left(\frac{I_0}{I} \right) = \epsilon cl$$

1. A solution of thickness 2cm transmits of 40% incident light. Calculate the concentration of the solution, given the molar absorptivity (ϵ) = 600 dm³/mol/cm.

$$A = 2 - \log_{10} (\%T)$$

$$A = 2 - \log_{10} 40 = 2 - 1.6020 \\ = \underline{0.398}$$

$$A = \epsilon l c$$

$$l = 2 \text{ cm}$$

$$\epsilon = 600 \text{ dm}^3/\text{mol}/\text{cm}$$

$$A = 0.398$$

$$c = ?$$

$$\therefore c = A/\epsilon l = 0.398 / 600 \times 2$$

$$= \underline{\underline{3.316 \times 10^{-5} \text{ mol/dm}^3}}$$

2) A soln shows a transmittance of 20%, when
passed in a cell of 2.5 cm thickness. Calculate its
concⁿ. If the molar absorption coefficient is
12000 dm³/mol/cm.

$$\rightarrow \underline{A = 2 - \log_{10} \%T} = 2 - \log_{10}(20) = 2 - 1.30 = 0.698$$

$$A = \epsilon \cdot c \cdot l$$

$$A = 0.698, \quad \underline{c = ?} \quad \epsilon = 12000 \text{ dm}^3/\text{mol}/\text{cm}.$$

$$l = 2.5 \text{ cm}$$

$$\therefore c = \frac{A}{\epsilon l} = \frac{0.698}{12000 \times 2.5} = \frac{0.698}{30000} = \underline{\underline{2.33 \times 10^{-5} \text{ mol/dm}^3}}$$

3) calculate molar absorptivity of a 1×10^{-4} M solution, which has an absorbance of 0.20, when the path length is 2.5 cm.

$$A = \epsilon c l$$

$$\cancel{L = 2.5 \text{ cm}}$$

$$A = 0.20$$

$$c = 1 \times 10^{-4} \text{ M}$$

$$l = 2.5$$

$$\therefore \epsilon = \frac{A}{c l}$$

$$= \frac{0.20}{1 \times 10^{-4} \times 2.5 \text{ cm}} = \frac{0.20}{0.00025} = 800 \text{ dm}^3/\text{mol}/\text{cm}$$

Fluorescence Spectroscopy

- **Fluorescence** is the absorption and emission of light of two different frequencies.
- Typically, a lower wavelength of incident light is absorbed from one direction, and a higher wavelength of light is emitted in all directions.
- Fluorescent molecules absorb light at a certain wavelength and emit at another.

What is fluorescence spectroscopy?

Fluorescence Spectroscopy analyzes fluorescence from molecules based on its fluorescent properties. Fluorescence is a type of luminescence caused by photons exciting a molecules raising it to an electronic excited state.

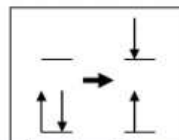
Principle of Fluorescence

1. Luminescence

- Emission of photons from electronically excited states → • Two types of luminescence:
 1. Fluorescence
 - Relaxation from singlet excited state
 2. Phosphorescence
 - Relaxation from triplet excited state

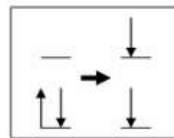
2. Singlet and triplet states

- Ground state – two electrons per orbital; electrons have opposite spin and are paired



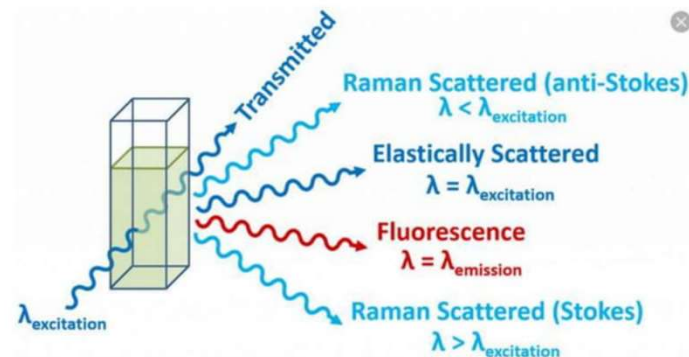
- Singlet excited state

Electron in higher energy orbital has the opposite spin orientation relative to electron in the lower orbital



- Triplet excited state

The excited valence electron may spontaneously reverse its spin (spin flip). This process is called intersystem crossing. Electrons in both orbitals now have same spin orientation

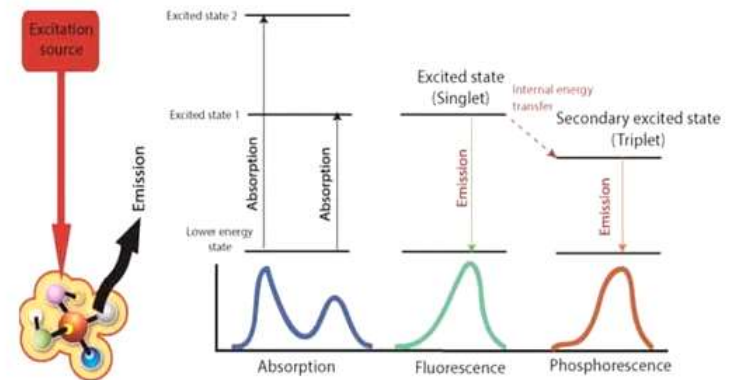
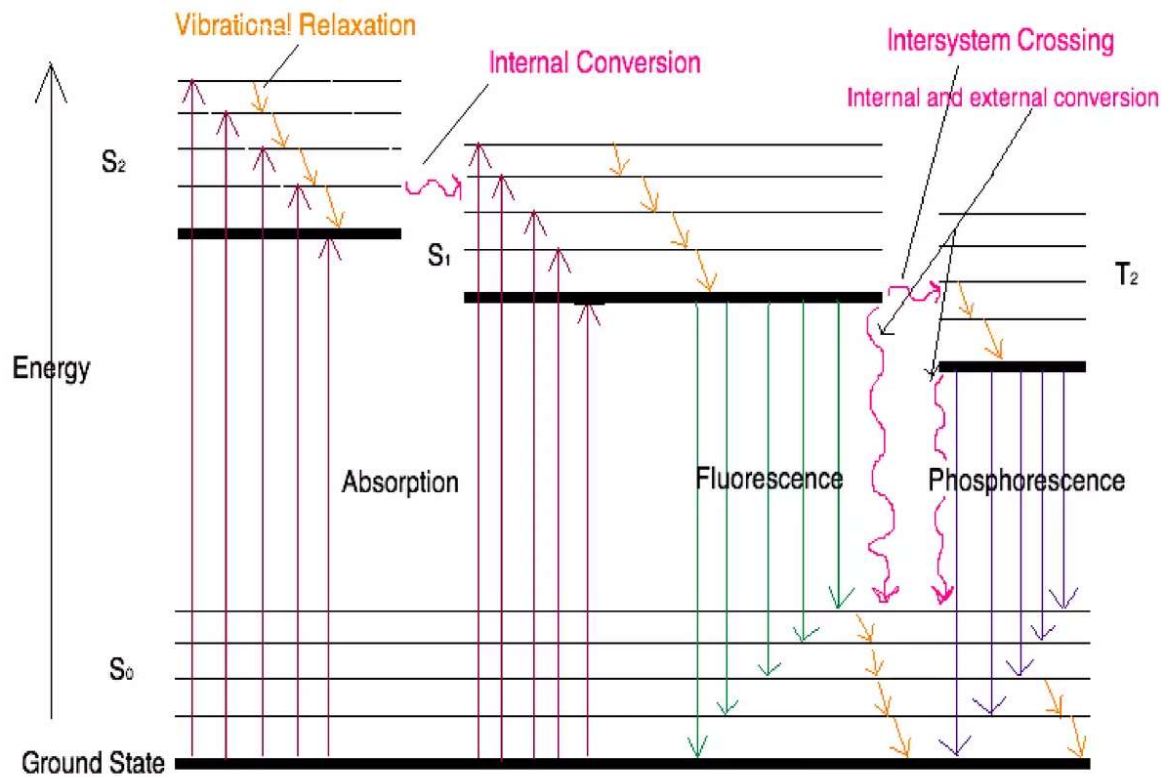


Relaxation processes

Once the molecule is excited to S_1 or S_2 several processes can occur that cause the molecule to lose its excess energy.

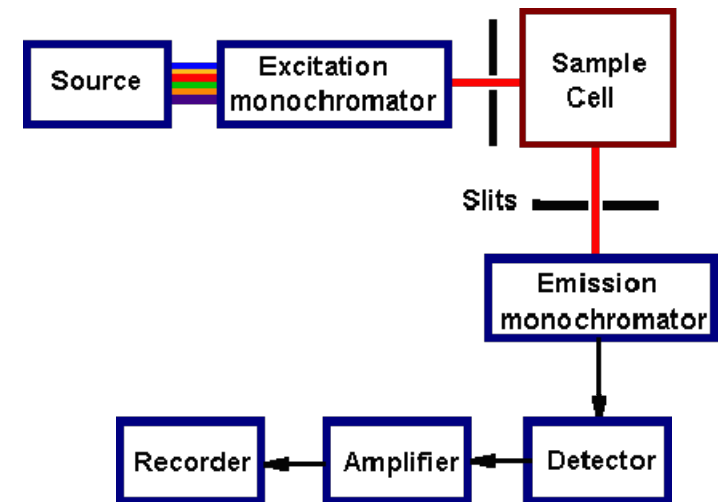
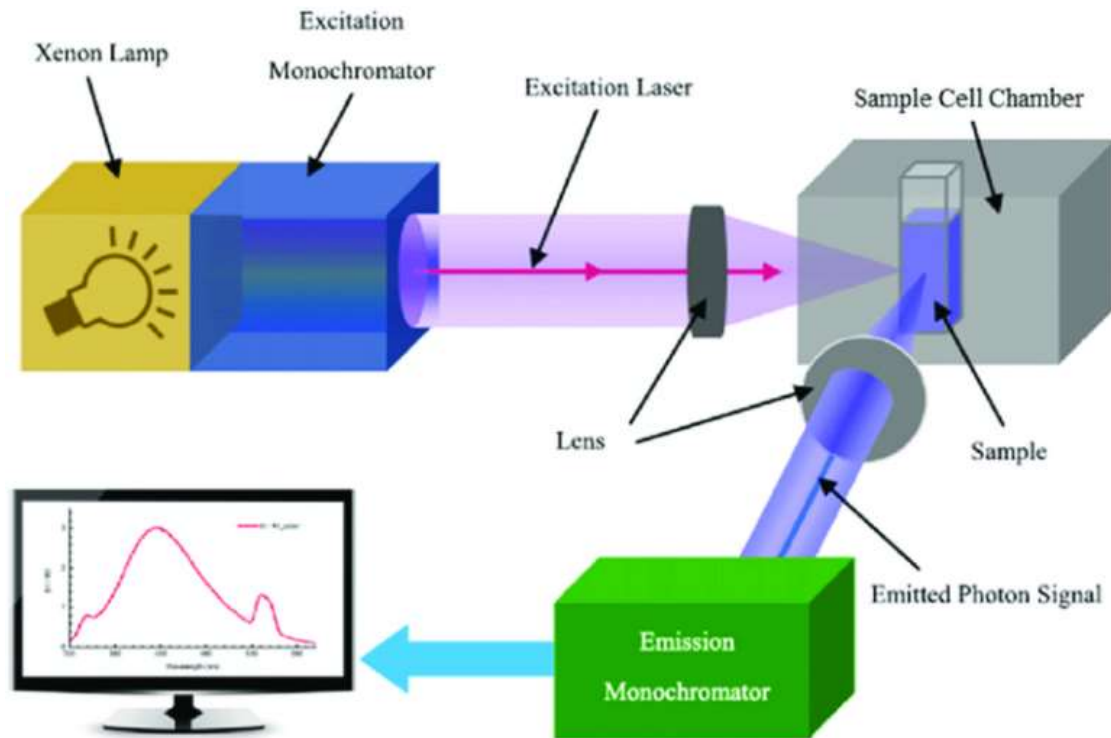
Various mechanism of relaxation:

1. Fluorescence
2. Phosphorescence
3. Vibrational relaxation
4. Internal conversion
5. External conversion
6. Inter system crossing



- **Fluorescence** – return from excited singlet state to ground state; does not require change in spin orientation (more common of relaxation)
- **Phosphorescence** – return from a triplet excited state to a ground state; electron requires change in spin orientation
- Emissive rates of fluorescence are several orders of magnitude faster than that of phosphorescence

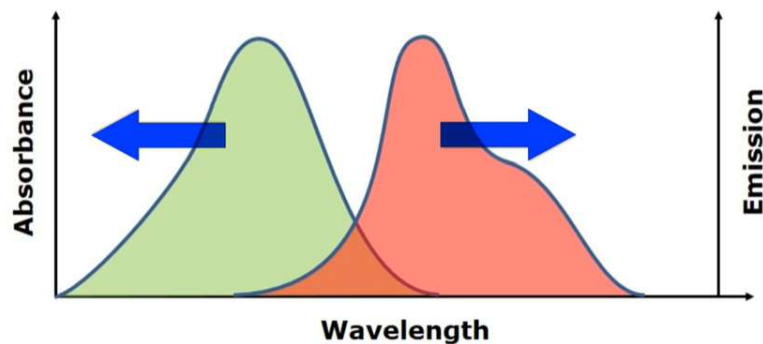
Instrumentation:



What is a Fluorescence Spectrum?

■ A fluorophore:

- Absorbs light at lower wavelengths (excitation)
- Emits light at longer wavelengths (emission)



The emitted light has a lower energy (lower frequency, longer wavelength) than the absorbed radiation; the difference in these energies is known as the **Stokes shift**.

“**Stokes shift** is the difference (in **energy**, **wavenumber** or **frequency units**) between positions of the band maxima of the **absorption** and **emission spectra** of the same electronic transition”

1. These two spectral types (emission and excitation) are used to see **how a sample is changing**.
2. The **spectral intensity** and or **peak wavelength** may change with variants such as **temperature**, **concentration**, or **interactions** with other molecules around it.
3. This includes **quencher molecules** (tethered molecule or a conjugated moiety) and molecules or materials that involve energy transfer.
4. Some fluorophores are also **sensitive to solvent environment** properties such as **pH**, **polarity**, and certain **ion concentrations**.

Applications

Fluorescence spectroscopy is often used in chemistry, biology, and environmental monitoring. Some examples:

1. **Industrial products**, for example in the petroleum industry, or pharmaceutical products can be monitored with detection of fluorescence, for example in order to detect impurities. One may identify substances and measure their concentrations.
2. **Environment monitoring**, e.g. on water samples or in air.
3. Various bacteria, viruses, fungi and parasites, causing dangerous infections, can be detected and distinguished through fluorescence spectroscopy.
4. **Cancer diagnostics in human tissues**. For example, one may infiltrate a sample with a substance which is preferentially accumulated in tumors and can then be detected through laser-induced fluorescence.
5. As already mentioned above, fluorescence markers are widely used in fluorescence microscopy for biological research.

NMR SPECTROSCOPY

INTRODUCTION

- Nuclear Magnetic Resonance (NMR) is a spectroscopy technique which is based on the absorption of electromagnetic radiation in the radio frequency region 4 to 900 MHz by nuclei of the atoms.
- Proton Nuclear magnetic resonance spectroscopy is one of the most powerful tools for elucidating the number of hydrogen or proton in the compound.
- It is used to study a wide variety of nuclei:

^1H

^{15}N

^{19}F

^{13}C

^{31}P

Theory of NMR:-

Spin quantum number (I) is related to the atomic and mass number of the nucleus.

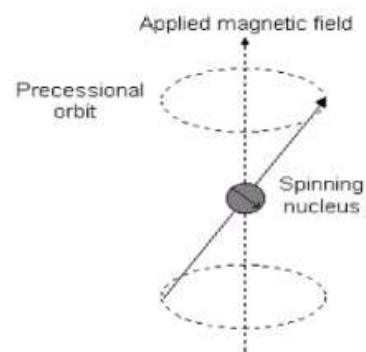
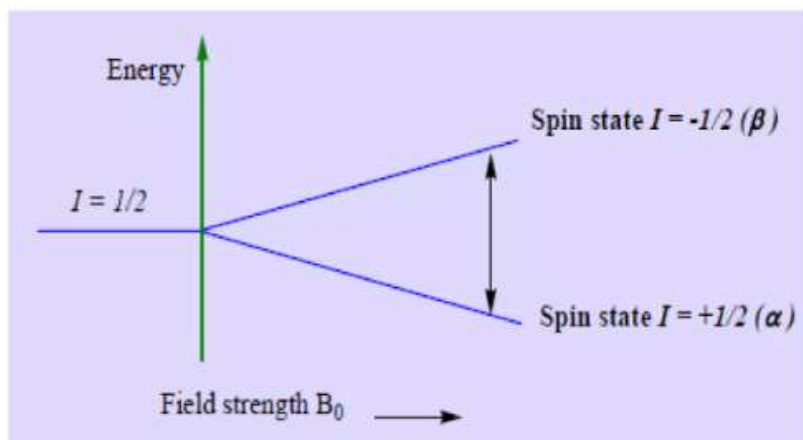
I	Atomic Mass	Atomic Number	Examples	
Half-integer	Odd	Odd	^1H ($1/2$)	NMR active
Half-integer	Odd	Even	^{13}C ($1/2$)	
Integer	Even	Odd	^2H (1)	
Zero	Even	Even	^{12}C (0)	Not NMR active

Elements with either **odd mass** or **odd atomic number** have the property of nuclear “spin”.

If an external magnetic field is applied, the number of possible orientations is calculated by $(2I+1)$.

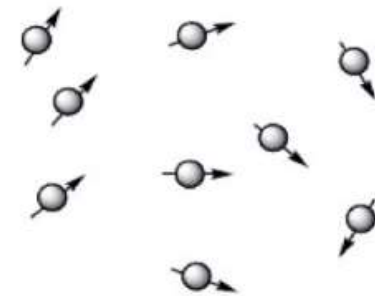
Hydrogen has spin quantum number $I=1/2$ and

possible orientation is $(2 \times 1/2 + 1 = 2)$ two $+1/2$ and $-1/2$.



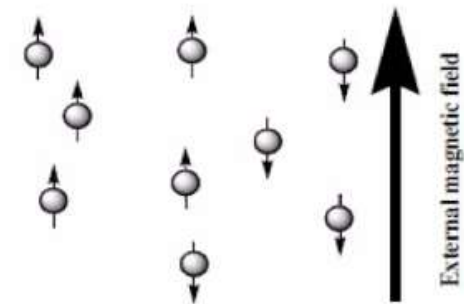
Principle of NMR

- The theory behind NMR comes from the spin of a nucleus and it generates a magnetic field.
- Without an external applied magnetic field, the nuclear spins are random in directions.



No external magnetic field

- But when an external magnetic field(B_0), is present the nuclei align themselves either with or against the field of the external magnet.

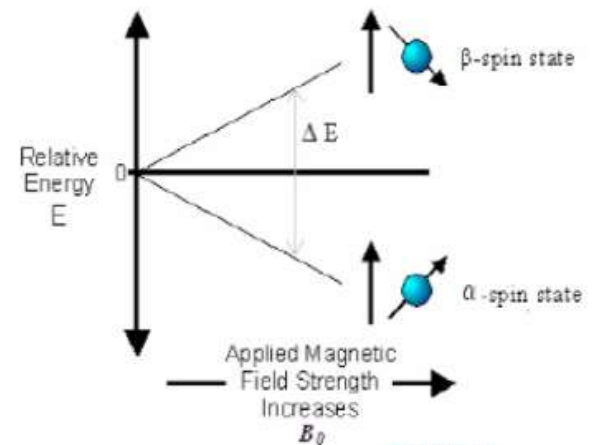
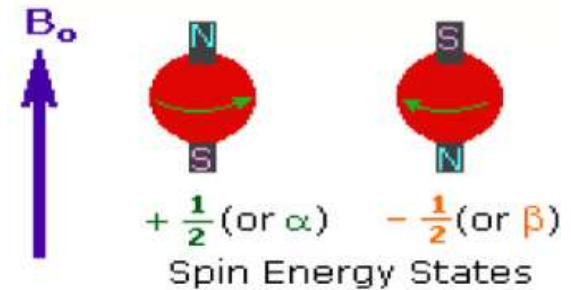


With external magnetic field

If an external magnetic field is applied, an energy transfer (ΔE) is possible between ground state to excited state.

when the spin returns to its ground state level, the absorbed radiofrequency energy is emitted at the same frequency level.

The emitted radiofrequency signal that give the NMR spectrum of the concerned nucleus.



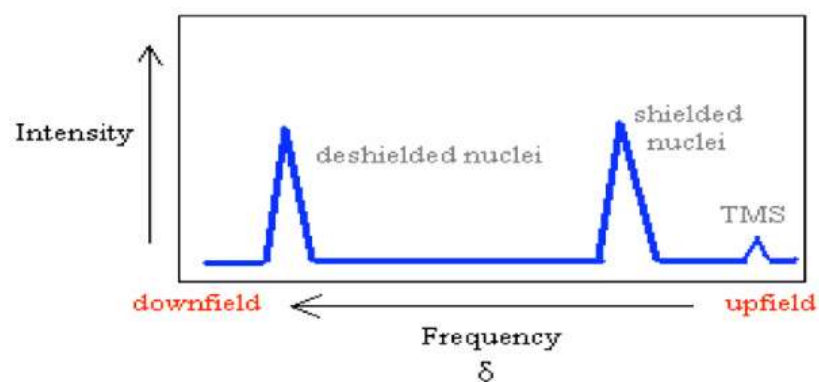
- The emitted radio frequency is directly proportional to the strength of the applied field.

$$\nu = \frac{\gamma B_0}{2\pi}$$

B_0 = External magnetic field experienced by proton

γ = Magnetogyric ratio (The ratio between the nuclear magnetic moment and angular moment)

NMR Spectrum



The NMR spectrum is a plot of intensity of NMR signals VS magnetic field (frequency) in reference to TMS.

Solvents used in NMR

The following solvents are normally used in which hydrogen replaced by deuterium.

CCl_4 - carbon tetrachloride

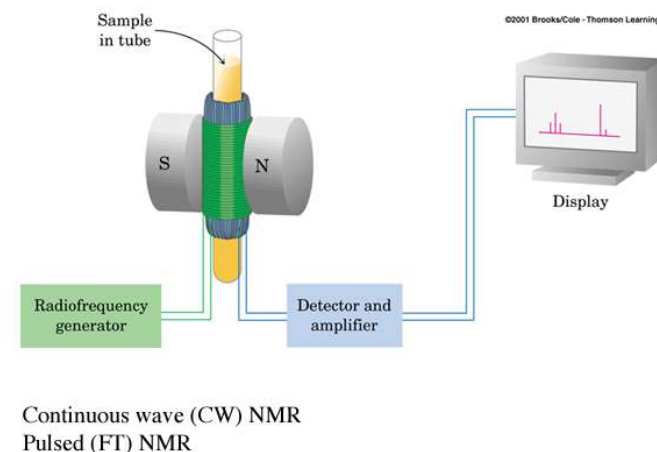
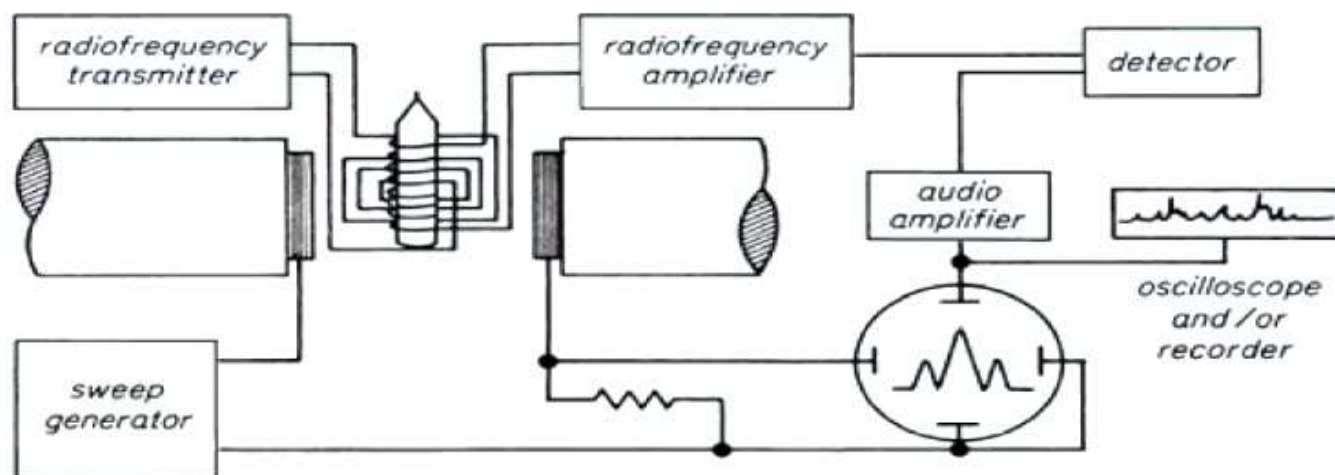
CS_2 - carbon disulfide

CDCl_3 - Deuteriochloroform

C_6D_6 - Hexa deuteriobenzene

D_2O - Deuterium oxide

NMR Instrumentation



1. **Sample holder:-** Glass tube with 8.5 cm long, 0.3 cm in diameter
2. **Permanent magnet:-** It provides a homogeneous magnetic field at 60-100 MHz
3. **Magnetic coils:-** These coils induce a magnetic field when current flows through them.
4. **Sweep generator:-** To produce an equal amount of magnetic field pass through the sample
5. **Radiofrequency:-** A radio transmitter coil transmitter that produces a short powerful pulse of radio waves
6. **Radiofrequency:-** A radio receiver coil that detects Receiver radio frequencies emitted as nuclei relax to a lower energy level
7. **Readout system:-** A computer that analyses and records the data.

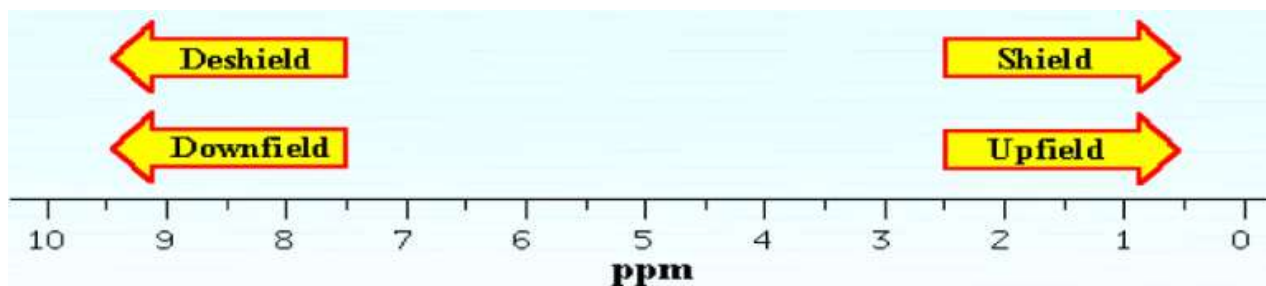
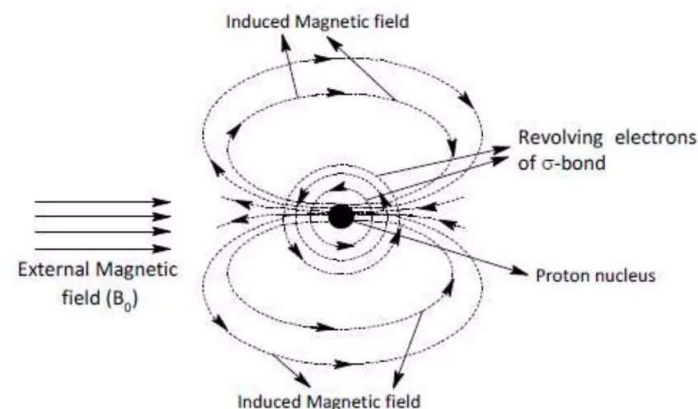
Shielding/Deshielding

shielding of protons:-

High electron density around a nucleus **shields** the nucleus from the external magnetic field and the signals are **upfield** in the NMR spectrum

Deshielding of protons:-

Lower electron density around a nucleus **deshields** the nucleus from the external magnetic field and the signals are **downfield** in the NMR spectrum



Chemical Shift

The location of an NMR signal in a spectrum for an organic compound relative to a reference signal (whose value is taken as zero) from a standard compound added to the sample is called as chemical shift of that compound. Such shifts (compared with standard reference) in the positions of NMR signals which arise due to shielding or deshielding of protons by electrons is called as chemical shift. Chemical Shift is having units of **parts-per-million** (ppm), and designated by the symbol δ . The value of chemical shift of an organic compound with respect to **TMS** can be defined as:

$$\delta = \frac{\nu_{\text{sample}} - \nu_{\text{reference}}}{\text{operating frequency in megacycles}}$$

$$\delta = \frac{\Delta\nu}{\text{operating frequency in megacycles}}$$

where ν_{sample} = Resonance frequency of the sample; $\nu_{\text{reference}}$ = Resonance frequency of TMS; $\Delta\nu$ = Frequency shift

Chemical Shift

The most of the organic compounds show chemical shift values between 0 and 10. In the τ -scale, the signal for TMS is taken as 10 ppm and the two scales are related to each other as follows:

$$\tau = 10 - \delta$$

NMR Applications

- 1) Widely used for structure elucidation.
- 2) Inorganic solids-inorganic compounds are investigated by solid state ^1H -NMR.
Eg: $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
- 3) Organic solids -solid ^1H NMR constitutes a powerful approach to investigate the hydrogen bonding and ionisation states of small organic compounds.
direct correlation with hydrogen bonding lengths could be demonstrated.
eg :for amino acid carboxyl groups.
- 4) Polymers and rubbers-examine hydrogen bonding and acidity
- 5) Peptides and proteins
- 6) In vivo NMR studies
 - ❖ concerned with ^1H NMR
 - ❖ spectroscopy of human brain
 - ❖ many studies are concerned with altered levels of metabolites in various brain diseases.
 - ❖ to determine the spatial distribution of any given metabolite detected spectroscopically .
- 7) Clinical and scientific research.

VIBRATIONAL SPECTROSCOPY (IR spectroscopy)

CONTENT

- Introduction
- Principle
- Type of vibrations
- Factors influencing vibrational frequency
- Instrumentation
- Application of IR spectroscopy

Introduction

- ❑ Infrared spectroscopy is one of the most important analytical technique used for determining the functional group present in both inorganic & organic compounds.
- ❑ Liquids, solutions, paste, powders, films, fibres, gases, and surfaces can all be examined with a choice of sampling technique.
- ❑ IR spectroscopy is a technique based on the vibrations of the atom of a molecule.
- ❑ An infrared spectrum is commonly obtained by passing infrared radiation through a sample and determining the fraction of the incident radiation which is absorbed at a particular energy.
- ❑ The term infra red covers the range of the electromagnetic spectrum between 0.78 and 1000 μm .

Infra red region has been divided into three sections

1. Near infra red region
2. Middle infra red region
3. Far infra red region

The wavelength ranges for the above regions are given below.

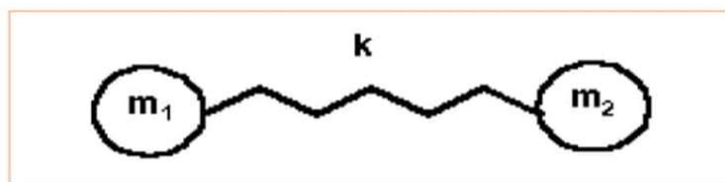
Region	Wavelength range (μm)	Wave number range (cm^{-1})
Near	0.78-2.5	12800-4000
Middle	2.5-50	4000-200
Far	50-1000	200-10

Thus **Hook's Law** states :

- The vibrational frequency is proportional to the strength of the spring; the stronger the spring; the higher the frequency.
- The vibrational frequency inversely proportional to the masses at the ends of the spring; the lighter the weights, the higher the frequency.

As per the **Hook's Law** :

- Stronger bonds absorb at higher frequencies.
- Weaker bonds absorb at lower frequencies.
- Bonds between lighter atoms absorb at higher frequencies.
- Bonds between heavier atoms absorb at lower frequencies.



$$\bar{\nu} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

$\bar{\nu}$ = frequency

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}$$

μ = reduced mass

- ❑ Consider a bond and the connected atoms to be a spring with two masses attached.
- ❑ Using the force constant **k** and the two masses **m₁** and **m₂** then the equation indicates how the frequency $\bar{\nu}$ of the absorption should change as the properties of the system change.
- ❑ The hooks law provides link between the strength of the covalent bond, masses of atoms and vibrational frequency.
- ❑ The greater the masses of attached atoms, the lower the IR frequency at which bond will absorb.

1. For a stronger bond (larger k value), μ increases :

As examples of this, in order of increasing bond strength compare:

- CC bonds: C–C (1000 cm^{-1}), C=C (1600 cm^{-1}) and C $\equiv$ C (2200 cm^{-1})
- CH bonds: C–C–H (2900 cm^{-1}), C=C–H (3100 cm^{-1}) and C $\equiv$ C–H (3300 cm^{-1})

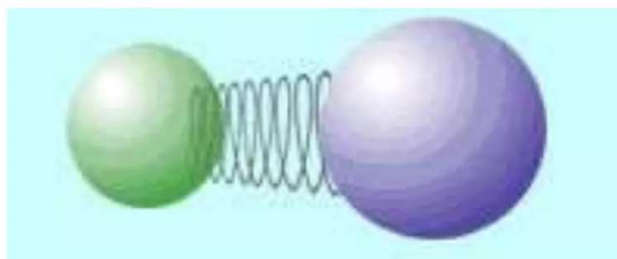
2. For heavier atoms attached (larger m value), μ decreases :

As examples of this, in order of increasing reduced mass compare:

- C–H (3000 cm^{-1})
- C–C (1000 cm^{-1})
- C–Cl (800 cm^{-1})
- C–Br (550 cm^{-1})
- C–I (about 500 cm^{-1})

Principle

- When energy in the form of infrared radiation is applied then it causes the vibration between the atoms of the molecules.
- When IR frequency is equal to the natural frequency of vibration absorption of IR radiation takes place, a peak is observed.
- IR spectrum of a chemical substance is a finger print of a molecule for its identification.



TYPES OF VIBRATIONS

- There are different types of vibrations:

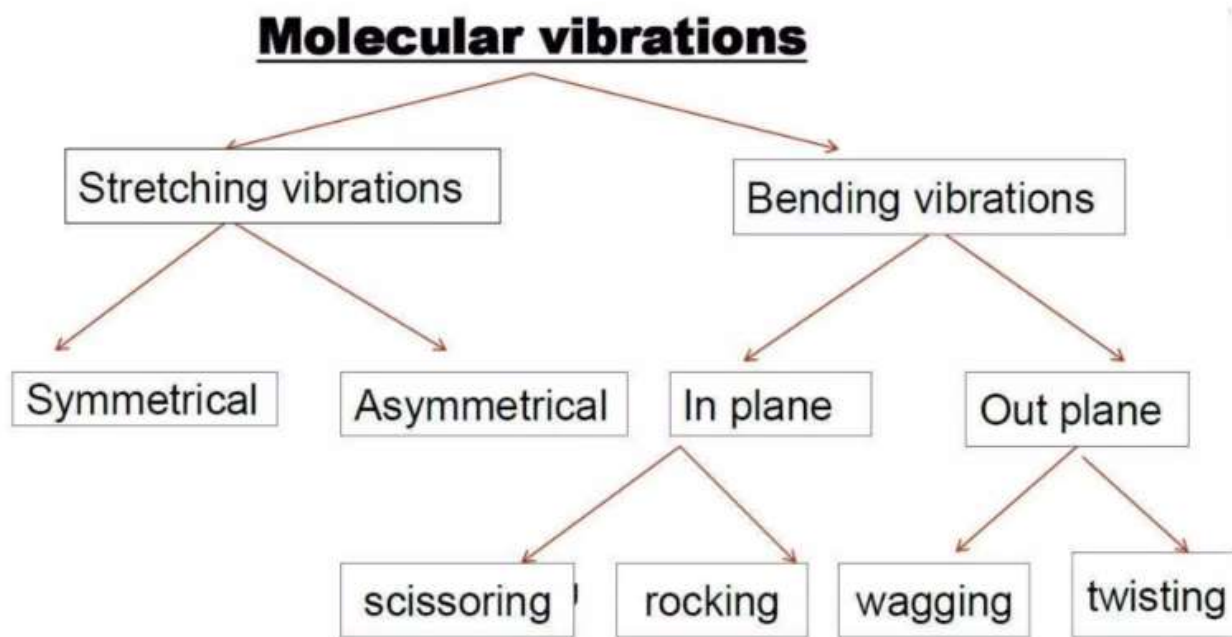
1. Stretching

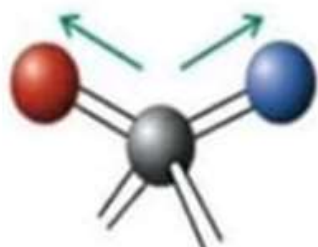
- i. Symmetric
- ii. Asymmetric

2. Bending

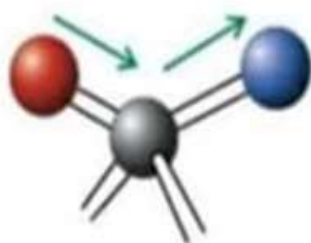
- i. In-plane bending
 - a. Scissoring
 - b. Rocking
- ii. Out-of-plane bending
 - a. Wagging
 - b. Twisting

Modes of molecular vibration

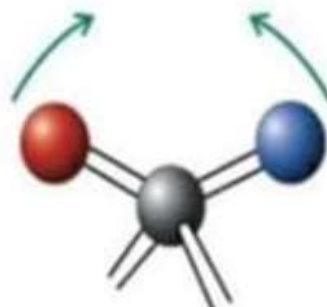




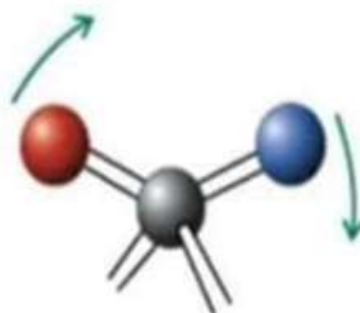
Symmetric stretching vibration (both outside atoms move away from or toward the center)



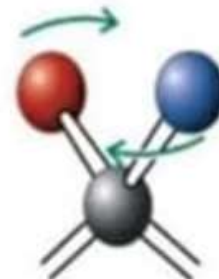
Asymmetric stretching vibration (as one atom moves toward the center, the other moves away)



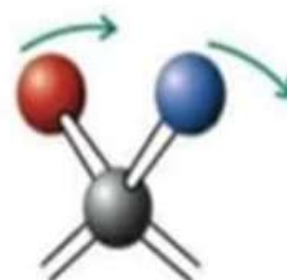
Symmetric bending vibration in a plane (scissoring)



Asymmetric bending vibration in a plane (rocking)



Symmetric bending vibration out of a plane (twisting)



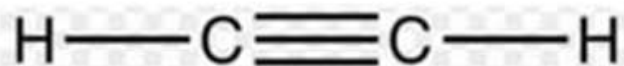
Asymmetric bending vibration out of a plane (wagging)

FACTORS INFLUENCING VIBRATIONAL FREQUENCY

1. Symmetry

Symmetric compounds do not possess dipole moment and are IR inactive.

E.g. symmetric acetylene



2. Fermi resonance

Fermi resonance results in an unexpected shift in energy and intensity of the bands.

E.g. the overtone of C-H deformation mode at 1400 cm^{-1} is always in Fermi resonance with the stretch of the same band at 2800 cm^{-1} .

3. Hydrogen bonding

Hydrogen bonding brings about remarkable downward frequency shifts.

Intermolecular – broad bands

Intra-molecular – sharp bands

4. Electronic effect

Electronic effects such as inductive, mesomeric and field effect may cause shift in absorption bands due to change in absorption frequency.

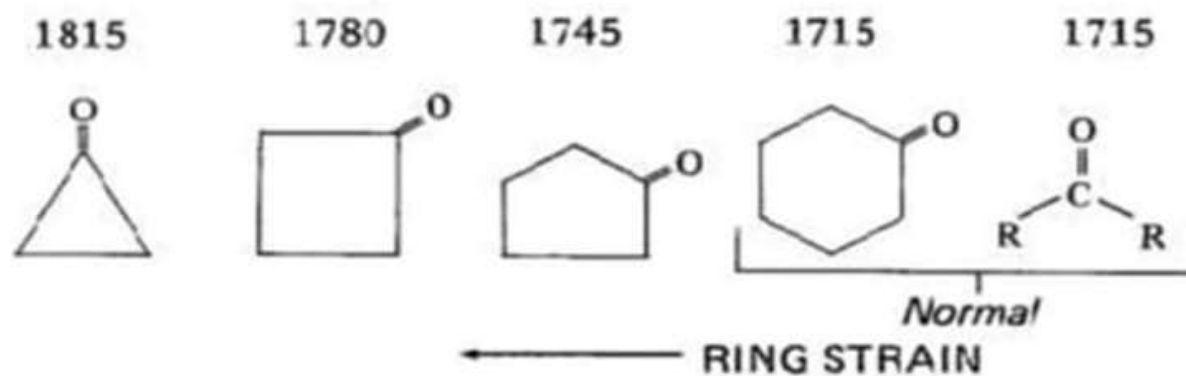
E.g. Inductive – acetone(1715cm^{-1}) and chloroacetone(1725cm^{-1})

Mesomeric- acetophenone(1693cm^{-1}) and p-amino acetophenone(1677cm^{-1})

5. Bond angles

Difference in bond angles also lead to the changes in absorption bands.

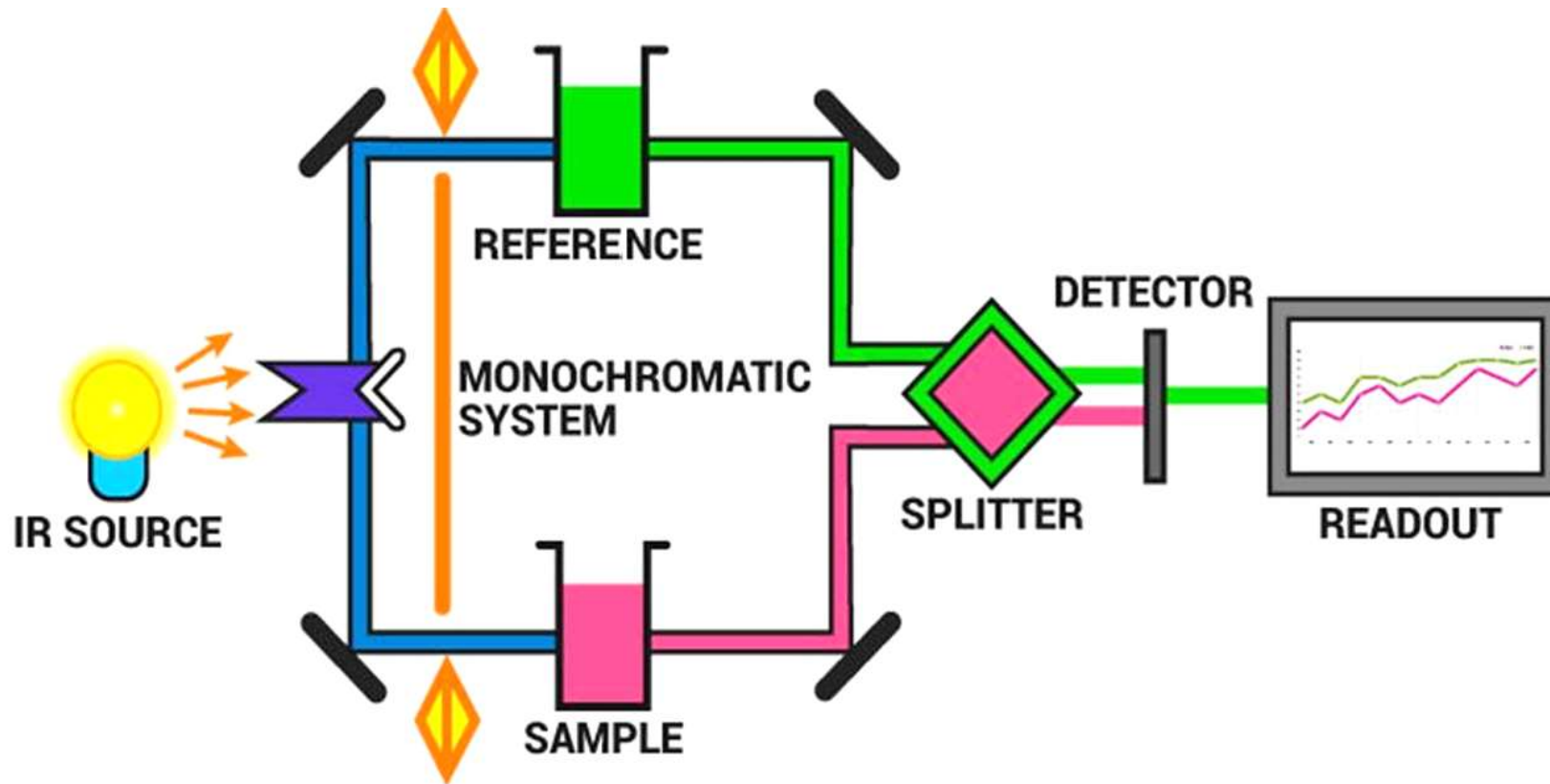
E.g.



INSTRUMENTATION

- The main parts of IR spectrometer are as follows :
- 1. IR radiation sources
- 2. Monochromators
- 3. Sample cells and sampling of substances
- 4. Detectors





- A beam of IR light from the source (Nernst Glower/Globar) is split into two and passed through the reference and the sample, respectively.
- Now, both of these beams are reflected to pass through a splitter and then through a detector. Finally, the required reading is printed out after the processor deciphers the data passed through the detector.

1. IR radiation sources

- The radiation source must emit IR radiation which must be
 - (i) intense enough for detection
 - (ii) steady
 - (iii) extend over the desired wavelengths
- The various popular sources of IR radiations are :
 - (i) Incandescent lamp
 - (ii) Nernst glower
 - (iii) Globar Source
 - (iv) Mercury Arc

RADIATION SOURCES

They must produce intense & steady radiation.

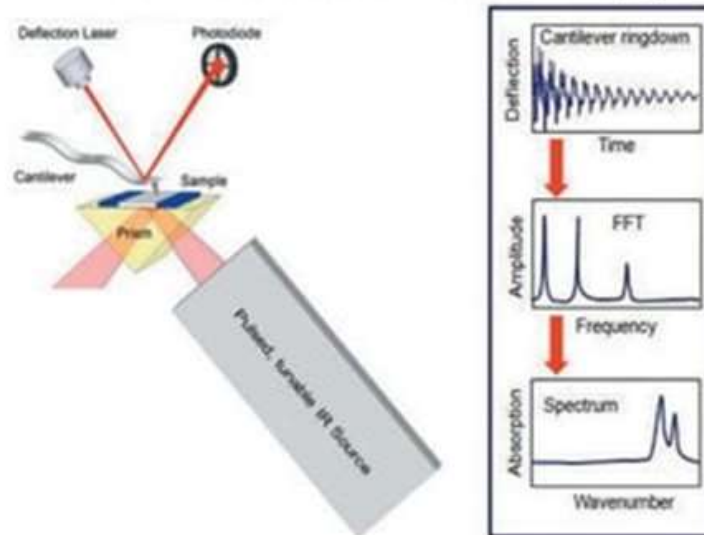


Nernst Glower	heated rare earth oxide rod (zirconium, yttria, thoria) (~1500 K)	1-50 μm (mid- to far-IR)
Globar	heated Silicon Carbide rod (~1500 K)	1-50 μm (mid- to far-IR)
W filament lamp	1100 K	0.78-2.5 μm (Near-IR)
Hg arc lamp	Hg plasma	50 - 300 μm (far-IR)

2. Monochromators

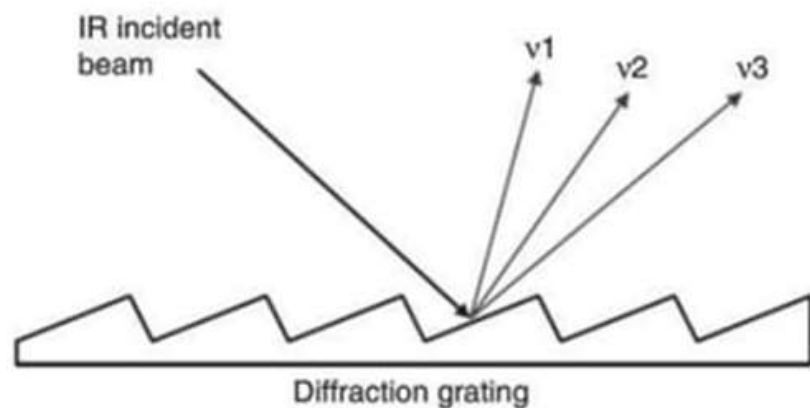
A. Prism:-

- Used as **dispersive element**.
- Constructed of various **metal halide** salts.
- **Sodium chloride** is most commonly prism salt used.



B. Grating

- Gratings are nothing but **rulings** made on some materials like glass, quartz or alkylhalides depending upon the instrument. The mechanism is that diffraction produces reinforcement. The rays which are incident upon the gratings gets reinforced with the reflected rays.



3. Sample cell & Sampling of substance

- Infrared spectra may be obtained for **gases, liquids or solids.**
- Materials containing sample must be **transparent** to the IR radiation. So, the salts like **NaCl** , **KBr** are only used.



Sample handling

- Samples of the same substance shows shift in absorption bands as we pass from solid to gases and hence the samples of different phases have to be treated differently in IR spectroscopy.
- **Sampling of solids**
 1. Solids run in solution
 2. Mull technique
 3. Pressed pellet technique
 4. Solids films

1. Solids run in solution

Dissolve solid sample in non -aqueous solvent and place a drop of this solution in alkali metal disc and allow to evaporate, leaving a thin film which is then mounted on a spectrometer.

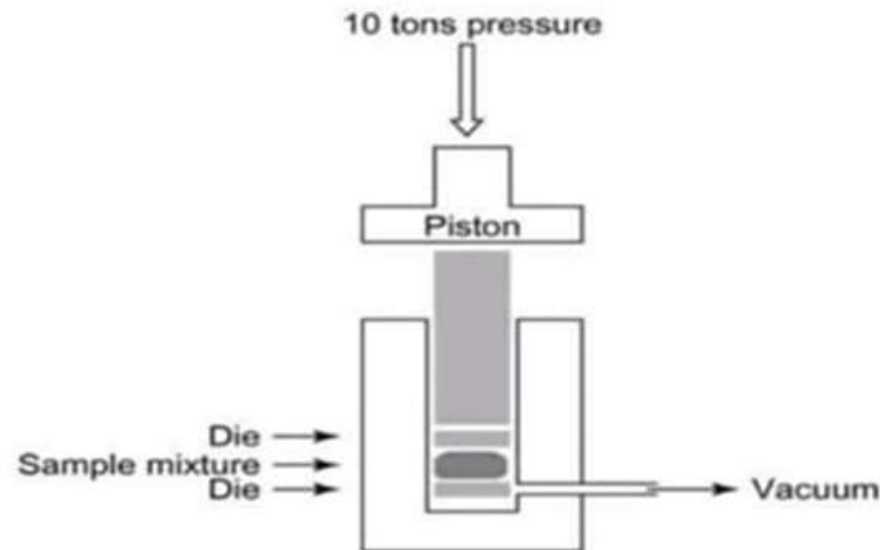
E.g. of solvents – acetone, cyclohexane, chloroform etc.

2. Mull technique

Finely powdered sample + mulling agent (Nujol) and make a thick paste (mull). Transfer the mull to the mull plates and the plates are squeezed together to adjust the thickness it is then mounted in spectrometer.

3. Pressed pellet technique

Finely powdered sample is mixed with about 100 times its weight of KBr in a vibrating ball mill and the mixture is then pressed under very high pressure in a die to form a small pellet (1-2mm thick and 1cm in diameter).



4. Solid films

Here amorphous solid is dissolved in volatile solvents and this solution is poured on a rock salt plate (NaCl or KBr), then the solvent is evaporated by gentle heating.

Sampling of gases

Here gases sample is introduced into a glass cell made up of NaCl.

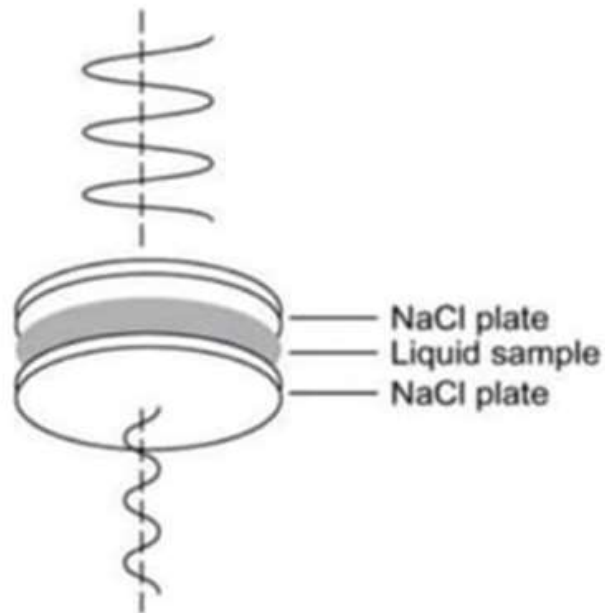
Very few organic compounds can be examined as gases.

E.g. : 1,4-dioxane

Sampling of liquids

Liquids sample can be sandwiched between two alkali halide plates (NaCl , KBr , CaF_2) .

The sample cell thickness is 0.01-0.05mm.



4. Detectors

- The detectors can be classified into three categories:

1. Thermal detectors:- Their responses depend the heating effect of radiation.

2. Pyroelectric detectors:- Pyroelectric effect depends on the rate of change of the detector temperature rather than on the temperature itself.

3. Photoconducting detectors:- Most sensitive.

Detectors

S.N	Name	Composed	Advantage
1.	Thermocouple	Thermoelectric effect - dissimilar metal junction	cheap, slow, insensitive
2.	Bolometer	Ni, Pt resistance thermometer (thermistor)	Highly sensitive <400 cm ⁻¹
3.	Golay cell	Metal cylinder with Xe gas	Faster than others & having wide wavelength range
4.	Pyro electric	Tri glycine sulfate piezoelectric material	fast and sensitive (mid IR)
5.	Photoconductive (non-thermal)	PbS, CdS, Pb Se light sensitive cells	fast and sensitive (near IR)

Dispersive IR instrument

- Dispersive IR instruments are introduced in 1940's.
- Double-beam instruments are mostly used than Single beam instrument.
- In dispersive IR sequential scanning of wave numbers of light takes place.



APPLICATIONS OF IR SPECTROSCOPY

- Identification of functional groups & structure elucidation of organic compounds.
- Quantitative analysis of a number of organic compounds.
- Study of covalent bonds in molecules.
- Studying the progress of reactions.
- Detection of impurities in a compound.
- Ratio of cis-trans isomers in a mixture of compounds.
- Shape of symmetry of an inorganic molecule.
- Study the presence of water in a sample.
- Measurement of paints and varnishes.

LIMITATION OF INFRARED SPECTROSCOPY

- By IR spectroscopy , it is not possible to know the molecular weight of a substance.
- It does not provide information of the relative positions of different functional groups on a molecule.
- From a single IR spectrum of an unknown substance, it is not possible to know whether it is a pure compound or a mixture of compounds.